

Counting Schur Rings Over the Cyclic Group of Order $4p$

Joseph Keller, Andrew Misseldine, Max Sullivan

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- $C_1 = 1$;
- for each i there is a j such that $C_i^* = C_j$;
- for each i and j we have $\overline{C_i} \cdot \overline{C_j} = \sum_{k=1}^r \lambda_{i,j,k} \overline{C_k}$, for constants $\lambda_{i,j,k} \in F$.

Schur Ring Vocab

Let $\mathfrak{S}$ be a Schur ring over G .

Definition (S-set)

An S-set is a union of primitive sets of a Schur Ring.

Definition (S-subgroup)

An S-set that forms a subgroup of G .

Definition (S-section)

Let K, H be S-subgroups of a Schur ring $\mathfrak{S}$. Then we say $[K, H]$ as an S-section if $1 \leq K \leq H \leq G$.

Trivial

Definition (Trivial Schur Ring)

A Schur ring is trivial if one class is the identity, and the other class is the rest of the group elements.

For any finite group, the trivial ring is always a Schur ring.

For any finite group G , the trivial Schur ring is always available, which we denote as G^0 .

Automorphic Schur Rings

Definition (Automorphic Schur Ring)

Let $\text{Aut}(G)$ be the automorphism group of G , and let $\mathcal{H} \leq \text{Aut}(G)$. Then $\mathcal{H}$ partitions G according to the automorphic action of $\mathcal{H}$ acting on G . This Schur ring is an *automorphic Schur ring* and is denoted $G^{\mathcal{H}}$.

In the special case that $\mathcal{H} = 1$, we simply denote G^1 as G , where all elements are isolated, the **Discrete Schur Ring**. For abelian

groups, let $G^{\pm}$ denote the automorphic Schur rings where elements come in inverse pairs, the **Symmetric Schur Ring**.

Direct Product Schur Rings

Definition (Direct Product Schur Ring)

Suppose G is a direct product of two groups, say $G = H \times K$. Let $\mathfrak{S}$ and $\mathfrak{T}$ be Schur rings over H and K , respectively. Then we can construct the **direct product** of Schur rings, denoted by $\mathfrak{S} \times \mathfrak{T}$, by the partition

$$\mathfrak{S} \times \mathfrak{T} = \text{Span}\{CD \mid C \in \mathfrak{S}, D \in \mathfrak{T}\}.$$

By construction, H and K are both $\mathfrak{S} \times \mathfrak{T}$ -subgroups.

Schur Quotient Rings

Definition (Schur Quotient Rings)

Let $\mathfrak{S}$ be a Schur ring over G and let K be a normal $\mathfrak{S}$ -subgroup of G . Let $\varphi : G \rightarrow G/K$ be the natural quotient map. Then $S/K = \varphi(\mathfrak{S})$ is a Schur ring over G/K , called a **Schur Quotient Ring**.

Of course, $(\mathfrak{S} \times \mathfrak{T})/H = \mathfrak{T}$ and $(\mathfrak{S} \times \mathfrak{T})/K = \mathfrak{S}$.

Wedge Products of Schur Rings

Definition (Wedge Decomposable)

A Schur ring $\mathfrak{S}$ is **wedge decomposable** if there is a proper $\mathfrak{S}$ -section $U = [K, H]$ such that for every primitive set C of $\mathfrak{S}$ either $C \subseteq H$ or C is a union of cosets of K . Otherwise, $\mathfrak{S}$ is **wedge-indecomposable**.

Wedge Products of Schur Rings

Definition (Wedge Product Schur Ring)

Let $U = [K, H]$ be a proper section of G , and let $\mathfrak{S}$ and $\mathfrak{T}$ be Schur rings over H and G/K , respectively. Suppose further that

- K is an $\mathfrak{S}$ -subgroup;
- H/K is a $\mathfrak{T}$ -subgroup;
- and $\mathfrak{S}/K = \mathfrak{T}_{H/K}$.

In this case, we define the **wedge product** of $\mathfrak{S}$ and $\mathfrak{T}$, denoted $\mathfrak{S} \wedge_U \mathfrak{T}$, by the rule $\mathfrak{S} \wedge_U \mathfrak{T} = \mathfrak{S} + \varphi^{-1}(\mathfrak{T})$, where $\varphi : G \rightarrow G/K$ is the natural map.

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Theorem (The Fundamental Theorem of Schur Rings over Cyclic Groups)

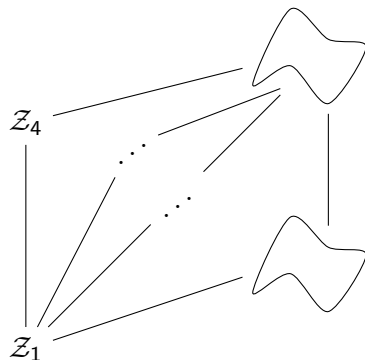
All Schur rings over cyclic groups are traditional [Leung and Man, 1996, Leung and Man, 1998].

Helpful Results

The important overlap that helps us get to our results are as follows:

- Direct products are Automorphic if and only if both products are Automorphic Schur rings.
- Direct products are Wedge products only if one of the products is wedge decomposable.
- Automorphic are wedge decomposable only if they map into the bottom layer of $\text{Aut}(\mathcal{Z}_{4p})$. [Misseldine, 2020]

Lattice of $\text{Aut}(\mathcal{Z}_{4p})$



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- an automorphic Schur ring which has $\mathcal{Z}_2$ and $\mathcal{Z}_p$ as an S-subgroup.

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- $\mathcal{Z}_p \wedge \mathcal{Z}_2$ which has $\mathcal{Z}_p$ as an S-subgroup.
- an automorphic Schur ring which has $\mathcal{Z}_2$ and $\mathcal{Z}_p$ as an S-subgroup.
- $\mathcal{Z}_{2p}^0$ which has only the trivial S-subgroup.

Wedges

In general, for any section $[a, b]$ the wedge product corresponding to it will be represented by $\mathfrak{S} \wedge_{\frac{b}{a}} \mathfrak{T}$ where $\mathfrak{S}$ is a schur ring over $\mathcal{Z}_b$ and $\mathfrak{T}$ is a Schur ring over $\mathcal{Z}_{\frac{4p}{a}}$.

Wedges II

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- $\mathfrak{S}$ over $\mathbb{Z}_p$ with section $[p, p]$ yields $\Omega(4, 1)\Omega(p, 1) = 3x$
- $\mathfrak{S} = \mathbb{Z}_4^0$ with section $[4, 4]$ yields $\Omega(p, 1) = x$
- $\mathfrak{S} = \mathbb{Z}_{2p}^0$ with section $[2p, 2p]$ yields $\Omega(2, 1) = 1$

Wedges III

- $\mathfrak{S} = \mathbb{Z}_4$ with section $[2, 4]$ yields $\Omega(2p, 2) = 2x$

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- $\mathfrak{S} = \mathcal{Z}_4$ with section $[2, 4]$ yields $\Omega(2p, 2) = 2x$
- $\mathfrak{S}$ automorphic over $\mathcal{Z}_{2p}$ with sections $[2, 2p]$, $[p, 2p]$, and $[2p, 2p]$ yields $\Omega(p) (2 + 2 - 1) = 3x$

Automorphic

Recalling our picture from a previous slide, the top layer includes the diagonals as well as the portion of the lattice in the upper right corner. This amounts to $\frac{2k+1}{k+1}x - 1$. However, the schur ring that maps to $\mathcal{Z}_2$ is $\mathcal{Z}_p^0 \times \mathcal{Z}_4$, which is automorphic since p a prime. So we have a total of $\frac{2k+1}{k+1}x$ automorphic Schur rings that are not wedge.

Direct

We need to consider direct products in which both are wedge indecomposable, and not both are automorphic. There is only one such form, which is $\mathcal{Z}_4^0 \times S$ where S is any schur ring over $\mathcal{Z}_p$. So there are x of these.

Counting formula

Theorem

For any odd prime $p = 2^k a + 1$ where a is an odd integer and x the number of divisors of $p - 1$

$$\Omega(4p) = \frac{15k + 14}{k + 1}x + 3$$

Where do we go?

- How does $\Omega(r, s)$ recurs?
- How do we generalize this to $\mathbb{Z}_{p^2q}$
- How do we generalize to any rank two?

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Thanks

Thank you for listening!